

RunZone: Full Project Guide

Geospatial Territory Conquest & Clinical Sports Science Endurance Platform

1 Executive Elevator Pitch (What is RunZone?)

💡 30-Second Quick Pitch

RunZone is a full-stack endurance running platform that transforms outdoor jogging into a real-time multiplayer territory conquest game while protecting athletes using clinical exercise physiology. Every GPS run generates a **40-meter geodesic corridor buffer in PostGIS** to capture and defend map sectors, while our **ACWR (Acute:Chronic Workload Ratio) engine** and **120-Billion Parameter AI Coach (ZoneCoach)** prevent soft-tissue injuries and prescribe calibrated workouts.

⚠️ The Industry Problem

Traditional running apps (Strava, Nike Run Club, Apple Fitness) are *passive data loggers*. They show static numbers after a run, offer no interactive territory game mechanics to motivate consistent habits, and lack real-time physiological guardrails to prevent overtraining injuries (which affect 50–70% of runners yearly).

🌟 The RunZone Solution

- 1. Gamification:** Conquering 40m buffered territory sectors across interactive Dark & Satellite maps.
- 2. Clinical Sports Science:** Dr. Tim Gabbett's ACWR model tracking 7-day fatigue vs 28-day chronic capacity.
- 3. Real-Time AI Coach:** Sub-second voice & chat inference powered by Groq GPT OSS 120B with health guardrails.

2 Complete Tech Stack & Architectural Breakdown

Layer	Technologies Used	Architectural Purpose & Engineering Rationale
Frontend	React 19, TypeScript, Vite, Tailwind CSS, Lucide Icons, Recharts	Ultra-fast client bundle (<400 kB gzip), strict type safety across all biometrics, responsive dark tactical UI aesthetic.
Map Engine	React-Leaflet v5, Leaflet v1.9, MapTiler Cloud, Esri, OSM	Zero-WebGL crash 2D map engine, MapTiler <code>streets-v2-dark</code> and <code>satellite-v2</code> raster layers with automatic Esri/OSM zero-key fallback.

Layer	Technologies Used	Architectural Purpose & Engineering Rationale
Backend API	Python 3.11, FastAPI, Pydantic v2, SQLAlchemy Async ORM	Asynchronous high-throughput REST API with automated OpenAPI docs, sub-15ms response times for auth, telemetry, and calculations.
Geospatial DB	PostgreSQL 16 + PostGIS extension, Shapely (Python)	Geodesic 40-meter corridor buffering (<code>ST_Buffer</code>), polygon union calculations, overlap conquest detection, and spatial indexing (GIST).
AI & Voice	Groq LPU (GPT OSS 120B, Whisper Large V3 Turbo)	Sub-second LLM inference for personalized coaching, hands-free voice transcription, and 3-tier clinical safety guardrails.
Cloud & CI/CD	Vercel (Frontend), Railway (Backend + DB), GitHub	Automated CI/CD pipelines, production environment variable isolation, and zero-downtime containerized deployments.

3 Core Technical Innovations (How It Works)

1. 40-Meter PostGIS Geodesic Corridor Buffering

When an athlete runs, their GPS trace is recorded as an array of coordinates `[[lon, lat], ...]`. In `spatial_service.py`, Shapely and PostGIS project these coordinates into planar meters using a geodesic latitude correction factor (`math.cos(math.radians(lat))`). A 20-meter radius on each side creates a **40-meter wide polygon corridor**. If this polygon intersects an existing sector, our algorithm calculates the captured area (km²), updates the faction ownership, and broadcasts the event via WebSockets.

2. Clinical ACWR (Acute:Chronic Workload Ratio) Safety Engine

Based on sports science research by Dr. Tim Gabbett (British Journal of Sports Medicine):

- **Acute Workload (Fatigue):** Exponentially weighted rolling average of TRIMP load over the last **7 days**.
- **Chronic Workload (Fitness Base):** Rolling average of TRIMP load over the last **28 days**.
- **ACWR Ratio:** $ACWR = \text{Acute Load} / \text{Chronic Load}$
- **Sweet Spot (0.80 - 1.30):** Optimal fitness adaptation with minimal soft-tissue injury risk (~10%).
- **Danger Zone (> 1.50):** Injury risk quadruples; the AI Coach automatically disables high-intensity workouts and enforces active recovery.

3. ZoneCoach 3-Tier Guardrail Safety Pipeline

Tier 1: Pre-Execution Interceptor

Catches emergency symptoms (chest pain, syncope), extracts chronic conditions (asthma, hypertension), and routes queries safely before calling the LLM.

Tier 2: Telemetry LLM Engine

Injects live Karvonen HR zones, 7-day mileage, ACWR, and medical limits into Groq GPT OSS 120B for scientific advice.

Tier 3: Post-Execution Validator

Inspects LLM output text. If the athlete's ACWR > 1.30 and the AI accidentally suggested sprints, the validator strips it and caps intensity at Zone 2.

4 Engineering Challenges & Solutions (STAR Stories)

Use these exact stories when the interviewer asks: "Tell me about a difficult technical problem you solved."

🔥 Challenge 1: Map Bleed-Through & Stacking Context Collision

Situation: When opening modal windows (like Route Planner or Activity Details), the background Leaflet map was bleeding through, intercepting click events and breaking the UI.

Task: Isolate the modal stacking context and ensure map controls never collide with overlay panels.

Action: Discovered that the modal was rendered inside a `<main className="overflow-y-auto">` container, creating an isolated CSS stacking context that trapped z-indices. Refactored the architecture by **lifting all modal state to the root level (`App.tsx`)** and elevated interactive HUD tactical controls to `z-[9000]` .

Result: Completely eliminated map bleed-through, achieving 100% modal isolation and smooth mobile/desktop interaction.

🔥 Challenge 2: AI Coach Repetitive Fallbacks & Zero-Downtime Resilience

Situation: Questions regarding 1600m splits, VO2 max intervals, and magnesium supplements were hitting fallback responses due to third-party API deprecations and strict regex filters.

Task: Build a resilient AI pipeline with dynamic model fallback and comprehensive sports science response matrices.

Action:

1. Queried the live Groq model catalog and migrated primary inference to `openai/gpt-oss-120b` .
2. Implemented dual-pass guardrail filtering (short conversational queries pass straight through to AI).
3. Built an offline clinical sports physiology matrix covering VO2 max, cadence (180 spm), magnesium glycinate, and respiratory regulation.

Result: 100% response uptime, sub-800ms generation speeds, and clinical-grade sports guidance in all online/offline conditions.

🔥 Challenge 3: Satellite Imagery Integration with Zero-Key Fallback

Situation: Needed photorealistic satellite basemaps across tactical maps without breaking existing dark streets, adding heavy map engines, or risking blank tiles if API limits were reached.

Task: Integrate MapTiler `satellite-v2` and Esri World Imagery fallback into React-Leaflet.

Action: Added a segmented HUD layer switcher control (Dark / Satellite) in `TerritoryMap.tsx` and `ActivityDetailModal.tsx` . Configured dual-tier tile resolution: MapTiler Cloud with automatic zero-key fallback to Esri World Imagery and OpenStreetMap.

Result: Users can seamlessly toggle between high-contrast dark tactical streets and high-resolution satellite imagery with zero latency and zero dependency additions.

5 Top 10 Technical Interview Questions & Perfect Answers

Q1: Why did you choose React-Leaflet over Mapbox GL or MapLibre?

"While Mapbox GL offers WebGL 3D vector tiles, it has a heavy bundle size (~1.5 MB), steep licensing costs, and frequent WebGL context crashes on low-end mobile devices during GPS background polling. **React-Leaflet v5 with Leaflet v1.9** is ultra-lightweight (~140 kB), renders raster tiles flawlessly at 60 FPS, and allows custom SVG/GeoJSON polygon manipulation with zero WebGL overhead."

Q2: How does the Acute:Chronic Workload Ratio (ACWR) calculate injury risk?

"ACWR compares the athlete's **Acute 7-day fatigue load** against their **Chronic 28-day fitness capacity**. We use TRIMP (Training Impulse), calculated from heart rate reserve and workout duration. When the ratio is between **0.80 and 1.30**, the athlete is in the physiological sweet spot. When it spikes above **1.50**, research shows a 4x increase in soft-tissue injury risk, so our AI Coach automatically adapts the training plan to active recovery."

Q3: How do you prevent the AI Coach from providing dangerous medical or training advice?

"We implemented a **3-Tier Safety Architecture**. Layer 1 screens for red-flag symptoms (e.g. chest pressure, fainting) and refuses to prescribe exercise. Layer 2 dynamically injects the athlete's chronic health conditions (e.g. cardiovascular history, asthma) and Karvonen heart rate zones into the LLM system prompt. Layer 3 is a deterministic post-execution validator that parses the AI's response and overrides any high-intensity recommendations if the athlete's ACWR is elevated."

Q4: Why did you choose FastAPI for the backend over Django or Express.js?

"FastAPI is built on **Starlette and Pydantic**, providing native `async/await` concurrency, automatic OpenAPI documentation, and strict schema validation. For geospatial and sports science calculations (Shapely, NumPy, PostGIS queries), Python is the industry standard. FastAPI provides sub-15ms execution while maintaining Python's rich scientific ecosystem."

Q5: How would you scale RunZone to support 100,000 active runners simultaneously?

"1. **Spatial Partitioning:** PostGIS spatial indexing with GIST and R-Tree partitioning to scope polygon intersection queries to local 10 km x 10 km bounding boxes.
2. **Redis Pub/Sub:** Offload real-time territory conquest WebSocket events from FastAPI to Redis clusters.
3. **Client-side Tile Caching:** Service Workers with Cache-First strategies for MapTiler and Esri raster tiles.
4. **Read Replicas & Connection Pooling:** Async SQLAlchemy with PgBouncer connection pooling to handle heavy read concurrency on leaderboards."

Q6: What makes RunZone unique compared to Strava or Nike Run Club?

"Strava focuses on retrospective social sharing and segment leaderboards. RunZone is an **active real-time territorial game and physiological guardian**. It gamifies every run into PostGIS territory conquest, prevents overtraining using clinical ACWR modeling, and provides an instant conversational sports science coach powered by high-speed Groq AI."

6 Quick Interview Cheat-Sheet Checklist

Key Metrics to Mention

- **40m:** Geodesic buffer width for captured corridors
- **0.80 – 1.30:** Gabbett ACWR sweet spot ratio
- **<800ms:** Groq GPT OSS 120B AI response speed
- **256px:** Native Leaflet raster tile scaling
- **15.0s:** Full production frontend build time

Top Keywords for Tech Screeners

PostGIS Geodesic Buffering · React-Leaflet v5 · Karvonen Heart Rate Reserve · TRIMP Workload Load · Gabbett ACWR Model · Groq 120B Inference · FastAPI Concurrency · CSS Stacking Contexts · Dual-Pass GPS Telemetry · Zero-Key Basemap Fallback